The Normal versus the T-Distribution

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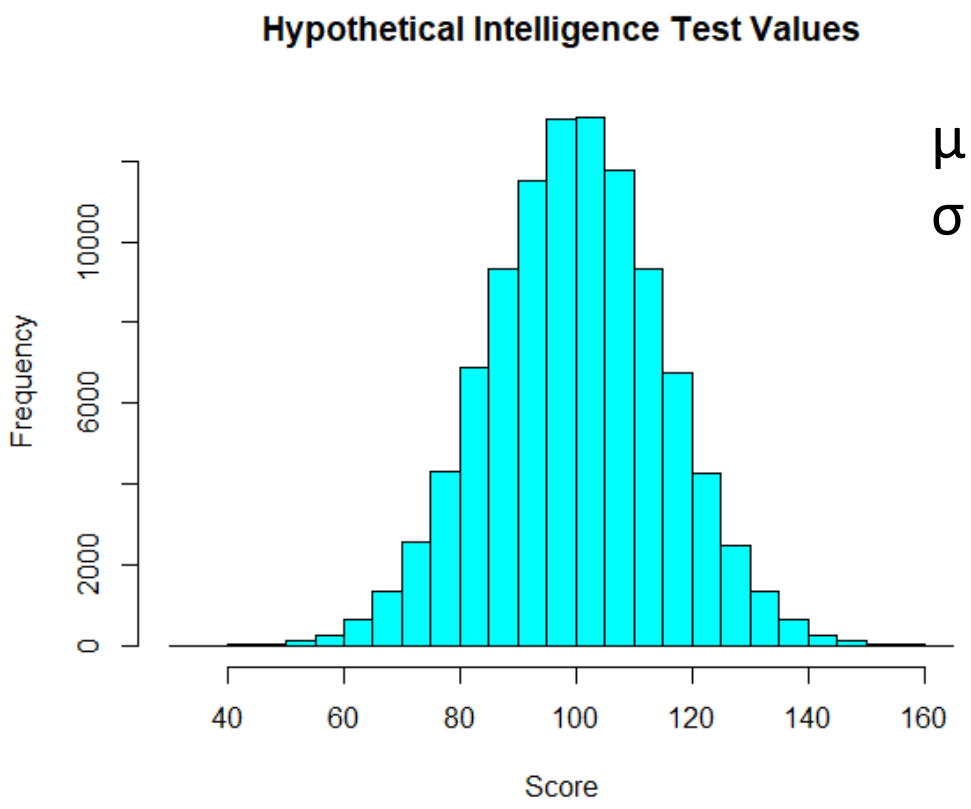
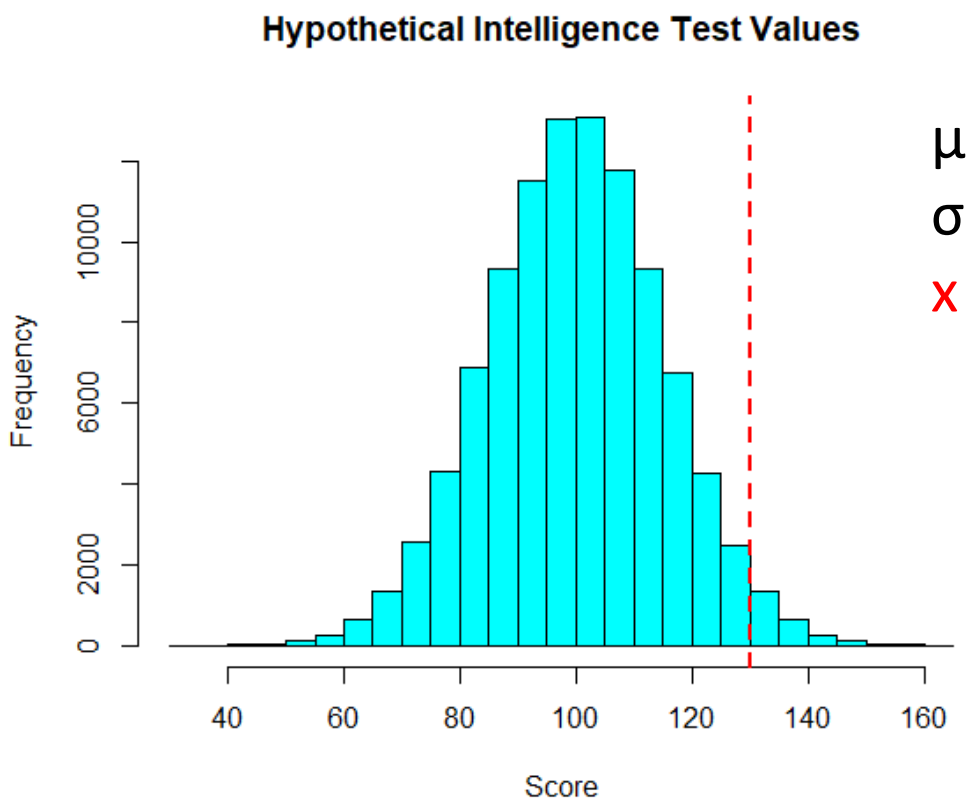
Finding the probability of an event in the normal distribution.



$\mu = 100$

$\sigma = 15$

Finding the probability of <u>an event</u> in the <u>normal</u> distribution.



$\mu = 100$

$\sigma = 15$

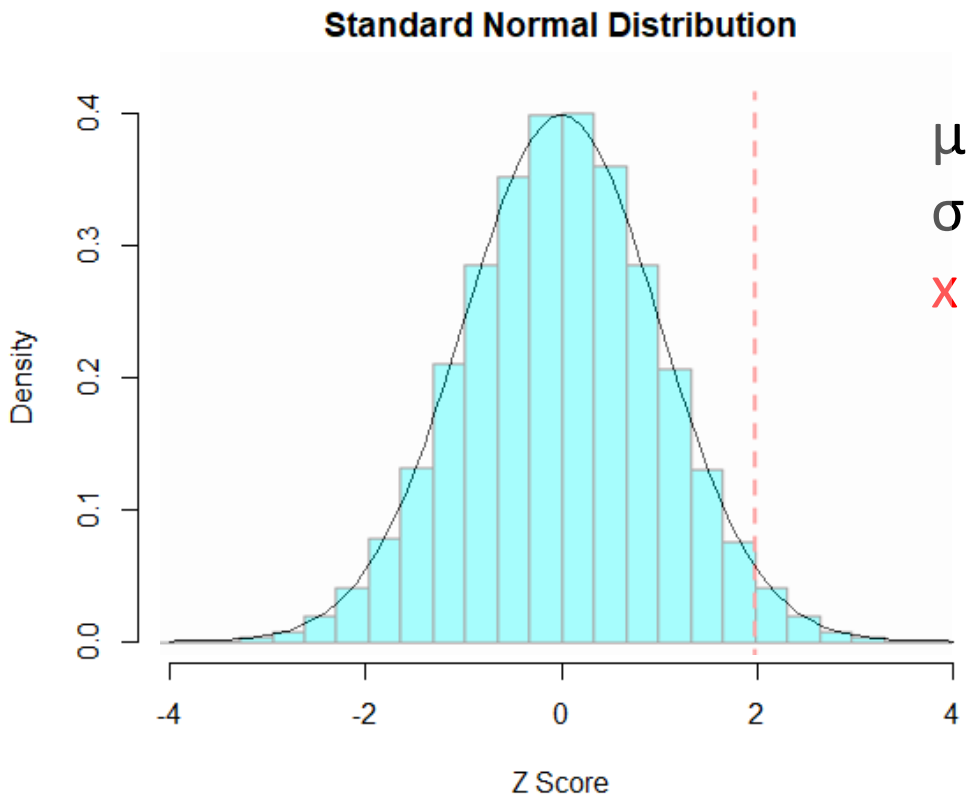
$x = 130$

What is $p(x \geq 130)$?

<u>Empirical</u> (if we have data): count the number of observations ≥ 130, divide by total number.

<u>Theoretical</u> (if we know the distribution): integrate to find the area under the curve ≥ 130, divide by total area.

Finding the probability of <u>an event</u> in the <u>normal</u> distribution.



μ = 100

σ = 15

x = 130

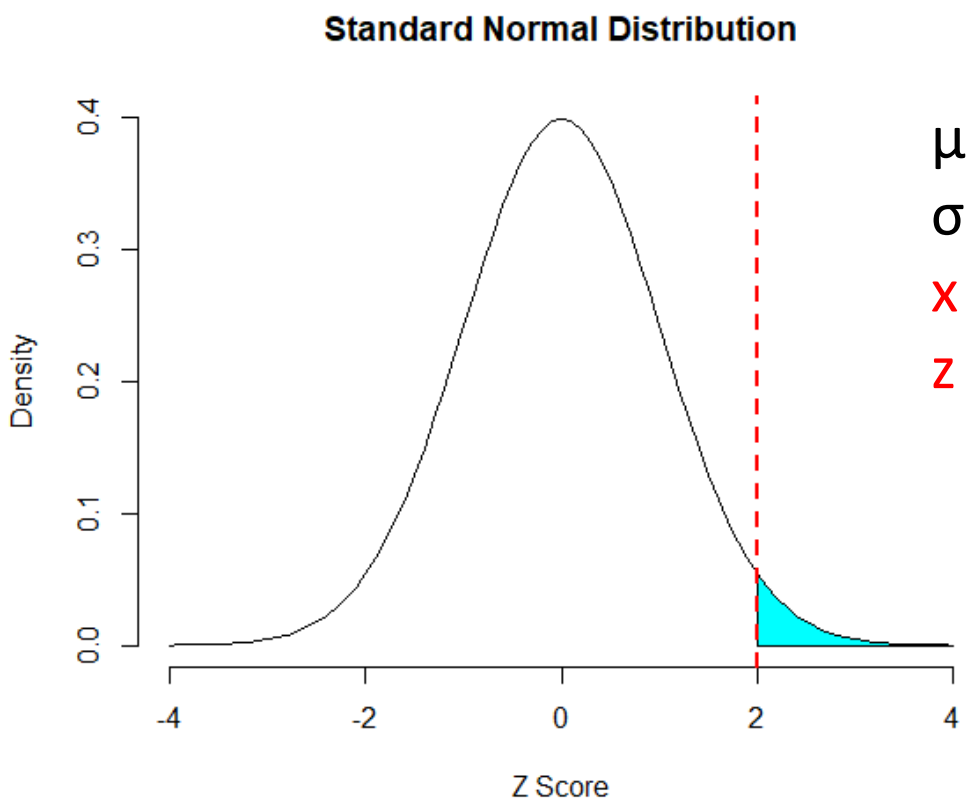
What is p(x ≥ 130)?

<u>Theoretical</u> (if we know the distribution): <u>integrate</u> to find the area under the curve ≥ 130, divide by total area.

$$z = \frac{(x - \mu)}{\sigma}$$

Solve for *pnorm(z)*

Finding the probability of <u>an event</u> in the <u>normal</u> distribution.



μ = 100
σ = 15
x = 130
z = 2.0

What is p(x ≥ 130)?

<u>Theoretical</u> (if we know the distribution): integrate to find the area under the curve ≥ 130, divide by total area.

$$z = \frac{(130 - 100)}{15} = 2.0$$

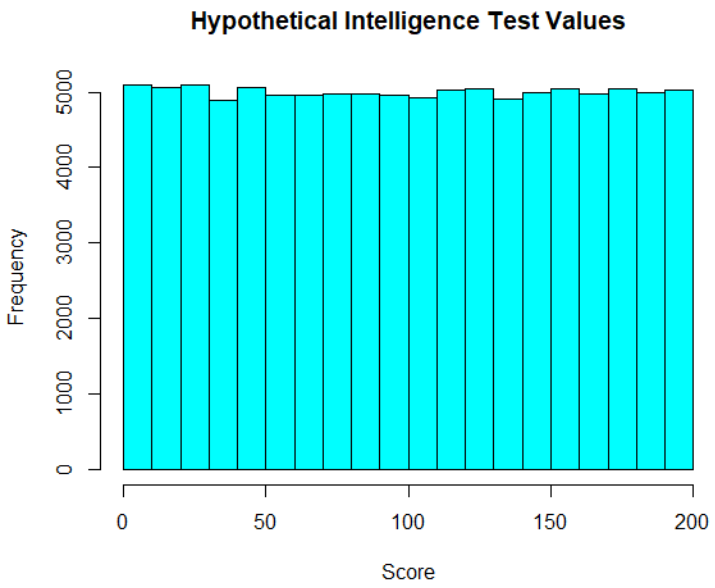
Solve for *pnorm(2.0)*
p(z ≥ 2.0) = 0.023

What about a sample mean?

- Recall from our previous video lectures that the distribution of sample means is **approximately** normal.
  - This is true even if the parent distribution is not normally distributed.
  - And as we showed, this distribution becomes more normal the larger our sample size gets.
  - This allows us to calculate the probability of observing a sample mean in a very useful way!

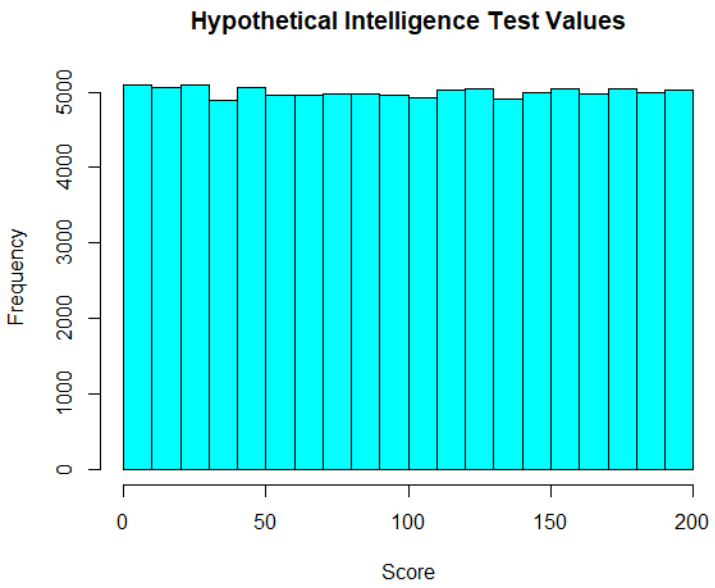
What about a sample mean?

Parent Population

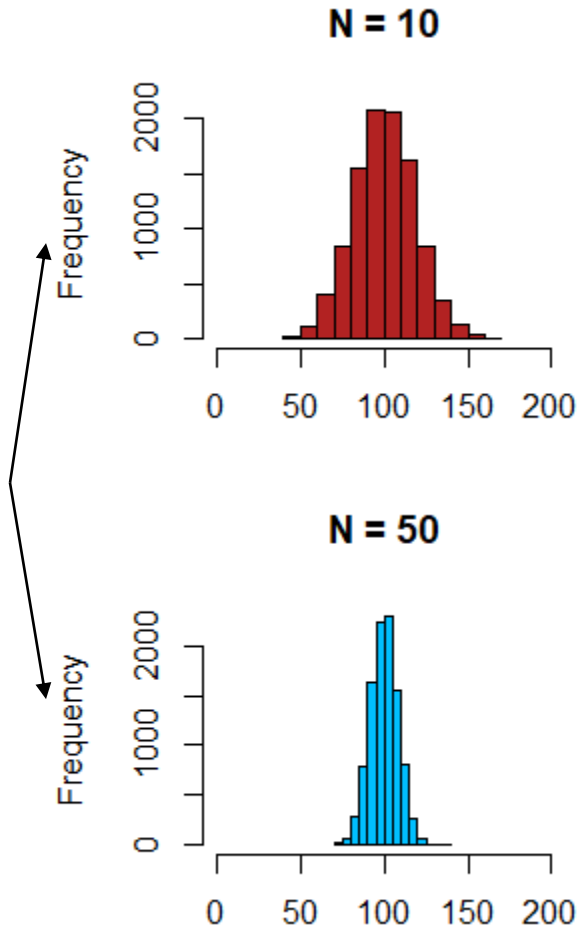


What about a sample mean?

Parent Population



Distributions of Sample Means



$$z = \frac{(x - \mu)}{\sigma}$$

The problem is that σ for the sampling distribution varies as a function sample size.

Specifically,

$$SE = \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{N}}$$

This special case of the standard deviation is referred to as the standard error of the mean (SE).

What about a sample mean?

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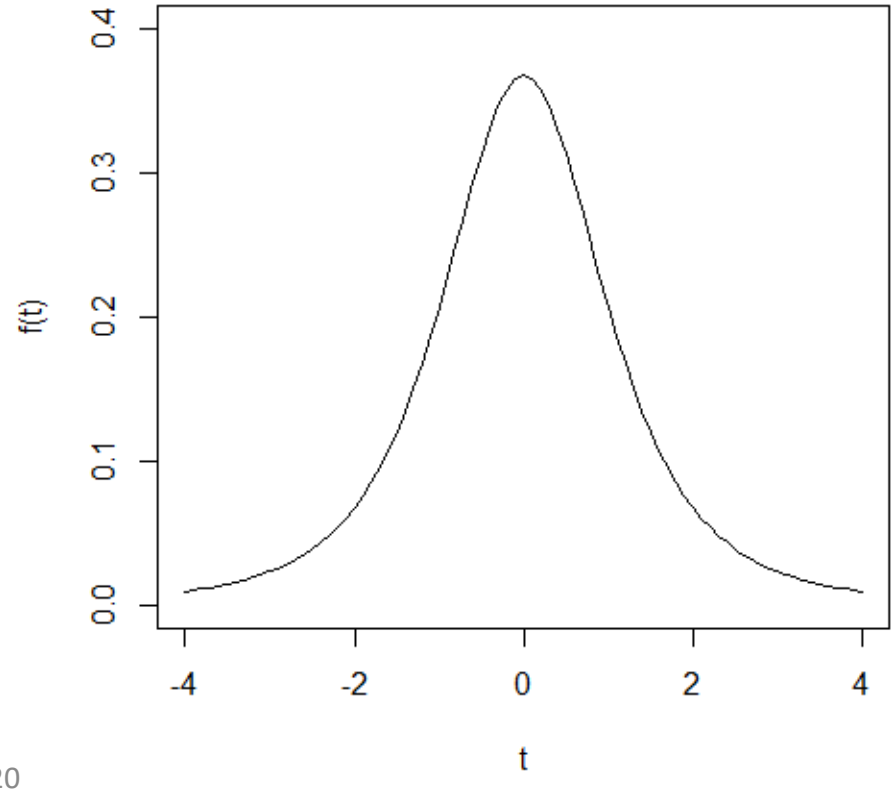

N = 3

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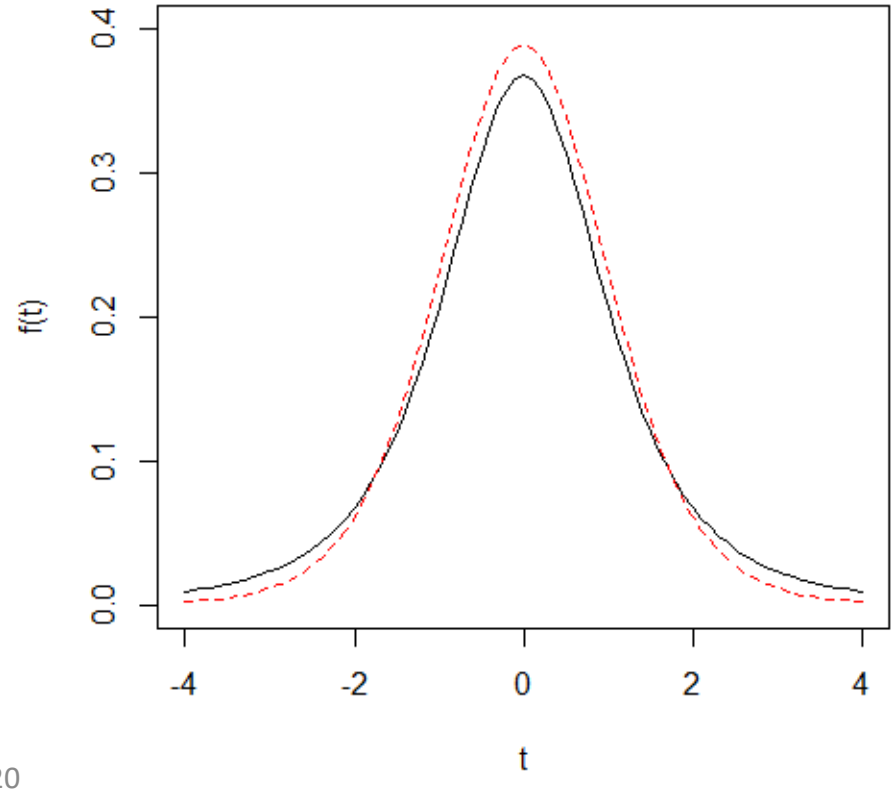
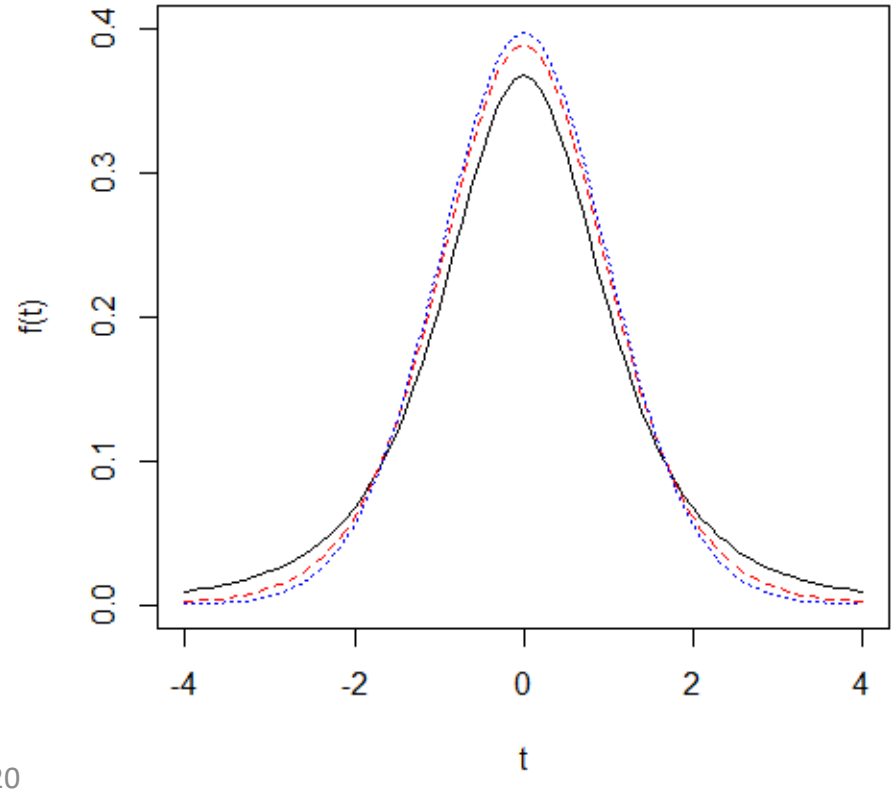


N = 10

N = 3

What about a sample mean?

- Recall from our previous video lectures that the distribution of sample means is **approximately** normal.



N = 50

N = 10

N = 3

What about a sample mean?

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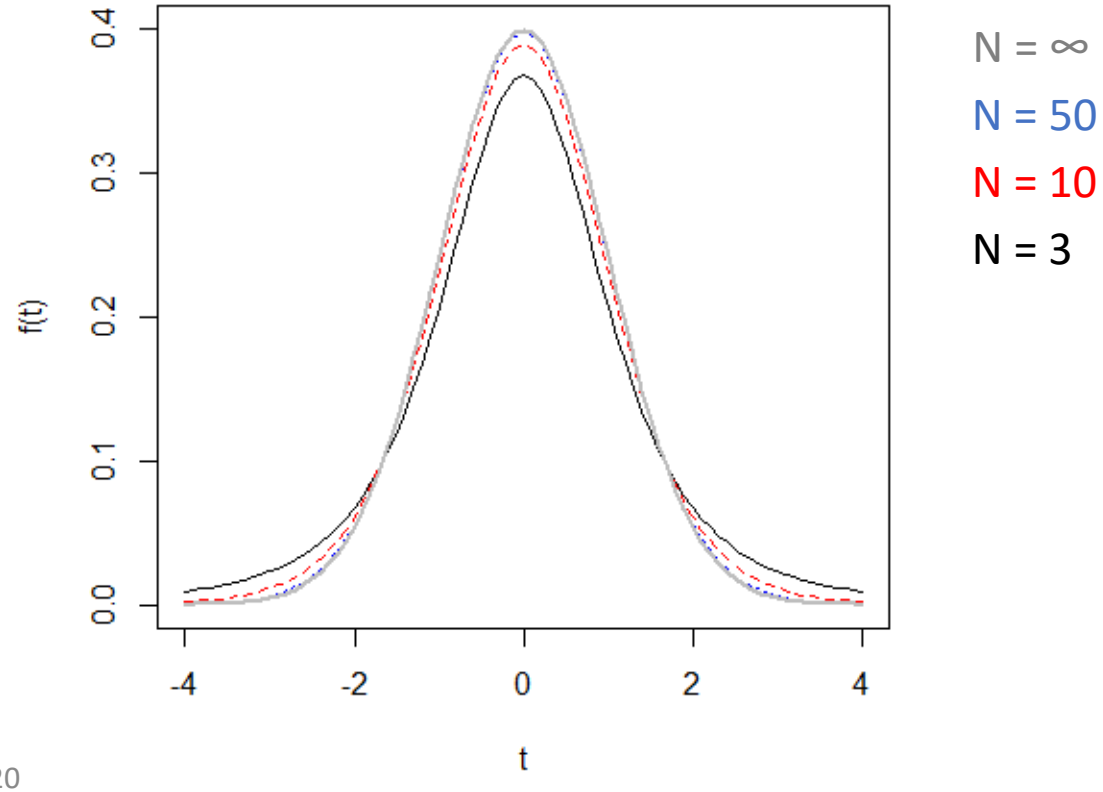

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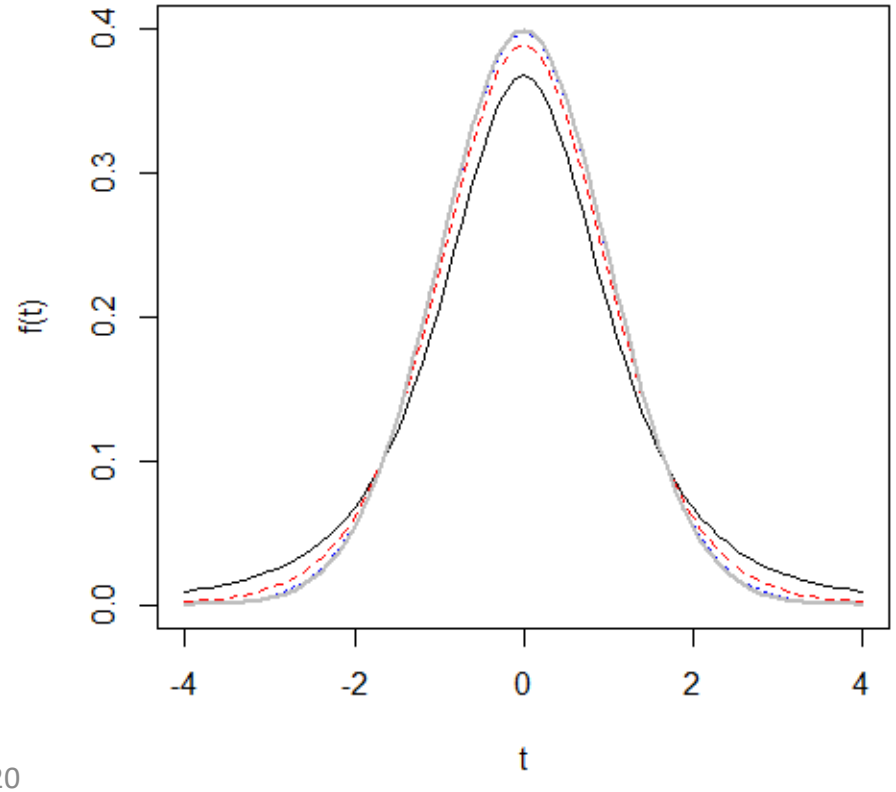

N = ∞

N = 50

N = 10

N = 3

This new distribution is the **t-distribution**.

It's shape is similar to the normal distribution, but varies based on number of observations.

As we calculated z-values, we can also calculate **t-values**, that tell us how far the sample mean is from the true mean in units of **standard error**.

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The major issue we will face, however, is that the population standard deviation (σ) is often unknown.

In those cases, we need to estimate σ using the standard deviation we observe in the sample (s).

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In lab, we will show that s is an unbiased estimate of σ if we divide by the appropriate degrees of freedom. For now, let's just use this formula.

What is probability of observing a sample mean in a t-distribution, N = 10.

μ = 100

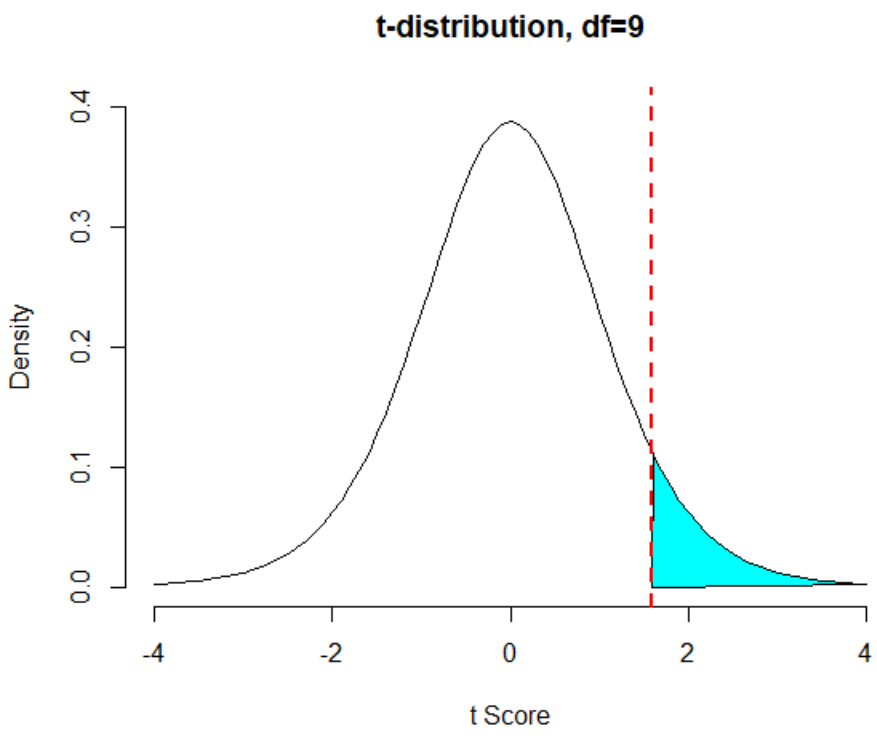
σ = ???

$\bar{x}$ = 115

s = 30

$$SE = \frac{30}{\sqrt{10}} = 9.487$$

$$t_{(9)} = 1.581$$



Solve for *pt(1.581, df=9)*

p* = *0.074

We would expect to observe a t-value of this size or greater about 7.4% of the time due to sampling variability alone, when the sample size is N=10.

What is probability of observing a sample mean in a t-distribution, N = 50.

μ = 100

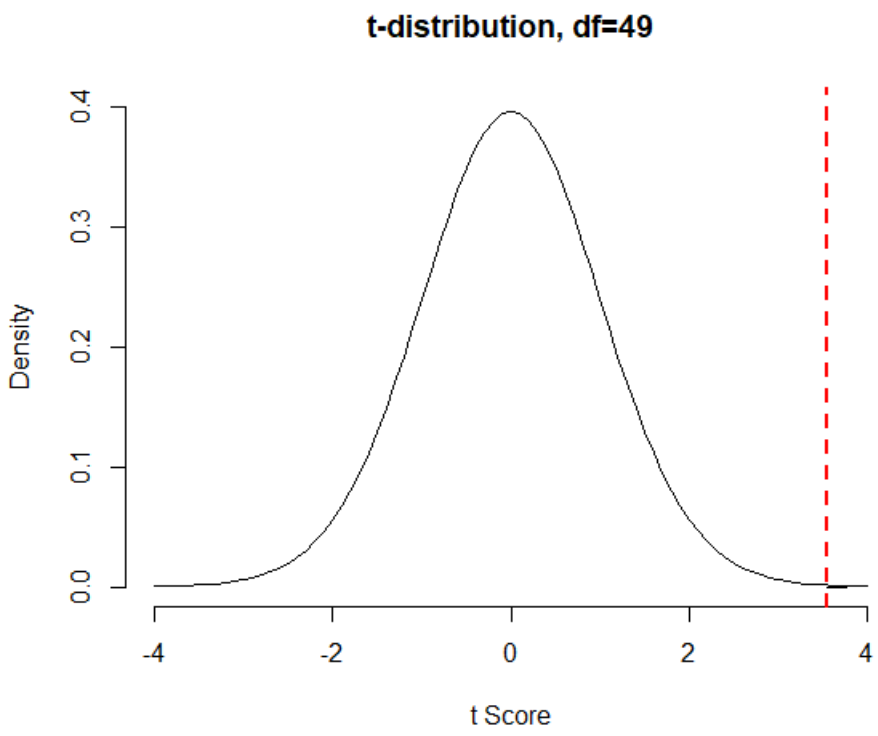
σ = ???

$\bar{x}$ = 115

s = 30

$$SE = \frac{30}{\sqrt{50}} = 4.243$$

$$t_{(49)} = 3.536$$



Solve for *pt(3.536, df=49)*

p* = *0.00045

We would expect to observe a t-value of this size or greater < 0.1% of the time due to sampling variability alone, when the sample size is N=50.

A generalized form of the t-statistic.

$$t_{obs} = \frac{(\hat{\beta} - \beta_H)}{SE_\beta}$$

The t-value we <u>observe</u> in our sample (because it would different in a different sample), tells us how far our estimated value is away from a hypothetical value, in units of standard error.

For instance, we might assume that the mean resting heart rate is 68 bpm for healthy young adults.

If we observe a mean heart rate of 60 bpm in a random sample of young adults, how likely is it that result occurred due to sampling variability alone?

A generalized form of the t-statistic.

$$t_{obs} = \frac{(60 - 68)}{s/\sqrt{N}}$$

It depends! ¯_(ツ)_/¯

How large is our sample? How variable is our sample?

The same difference could lead us to very different conclusions.

For instance, if we assume s = 12 and n = 5, then $t_{obs} = -1.49$.

But if we assume s = 12 and n = 20, then $t_{obs} = -2.98$.

Thus, the "surprise-value" of an effect depends on the size of the effect, the variability of the data, and the size of our sample.